

Customer Review Sentiment Analysis

AI-Powered Sentiment Classification on Amazon Reviews 2023 (TF-IDF + Logistic Regression & DistilBERT)

Project Report — Data Quality, EDA, Preprocessing, Feature Engineering, Modeling & Business Insight

235,144

reviews analysed

83.0%

positive sentiment

0.605

baseline macro-F1

0.888

baseline ROC-AUC

235,144 reviews across 5 product categories — 1997-09-10 to 2023-03-18

1. Business Problem

Online retailers receive customer reviews far faster than any team can read them. The business needs to automatically gauge **what customers feel** about products, surface **why** they are dissatisfied, and route problem products for action — without manual triage.

Business questions

- What share of reviews are positive, neutral or negative?
- Which product categories generate the most negative sentiment?
- What words and phrases drive negative reviews?
- Can sentiment be predicted accurately enough to automate monitoring?

Approach

Each review is classified into **negative / neutral / positive** using star ratings as labels (1–2★ negative, 3★ neutral, 4–5★ positive). Two models are compared: an interpretable **TF-IDF + Logistic Regression** baseline and a fine-tuned **DistilBERT** transformer.

2. Dataset & Data Quality

The project uses the **Amazon Reviews 2023** corpus, sampled across **5 product categories**. The raw sample was cleaned and de-duplicated before labeling.

Metric	Value
Raw rows downloaded	250,000
Rows dropped (missing text)	51
Duplicate reviews removed	14,805
Final clean reviews	235,144
Binary subset (3★; dropped)	216,895
Verified purchases	65.27%
Mean helpful votes / review	1.74
Review date range	1997-09-10 to 2023-03-18

2.1 Reviews per category

Category	Reviews
Home and Kitchen	47,409
Electronics	47,370
Clothing Shoes and Jewelry	47,020
Books	46,980
Toys and Games	46,365

3. Exploratory Data Analysis

3.1 Rating & sentiment distribution

The corpus is heavily positive-skewed: **83.0% positive**, **7.8% neutral**, **9.2% negative**. This class imbalance is the central modeling challenge and is why the baseline uses balanced class weights and reports macro-F1 alongside accuracy.

Star ratings are dominated by 5★ reviews (154,933), with relatively few 1–2★ reviews — mirroring the sentiment skew.

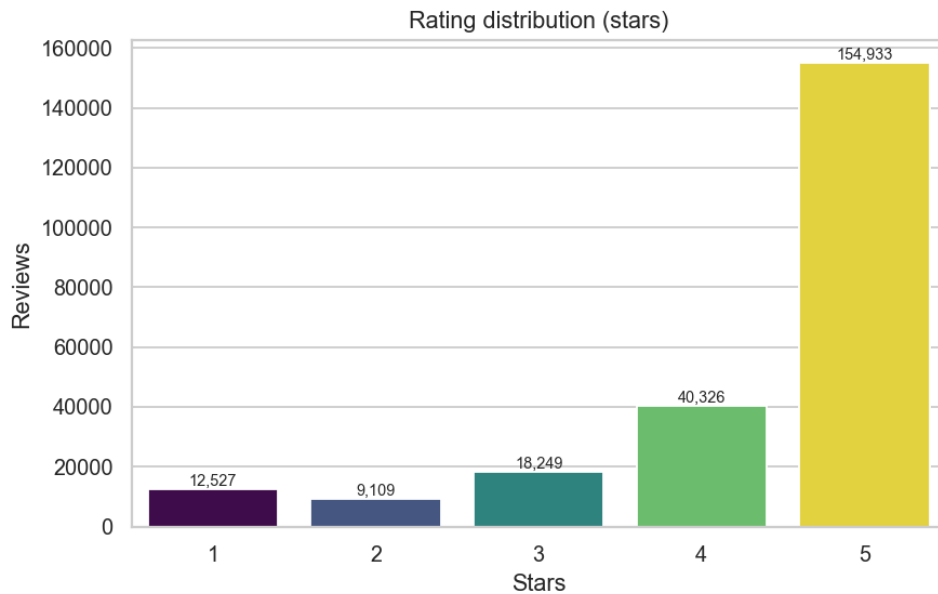


Fig 1. Distribution of star ratings (1–5).

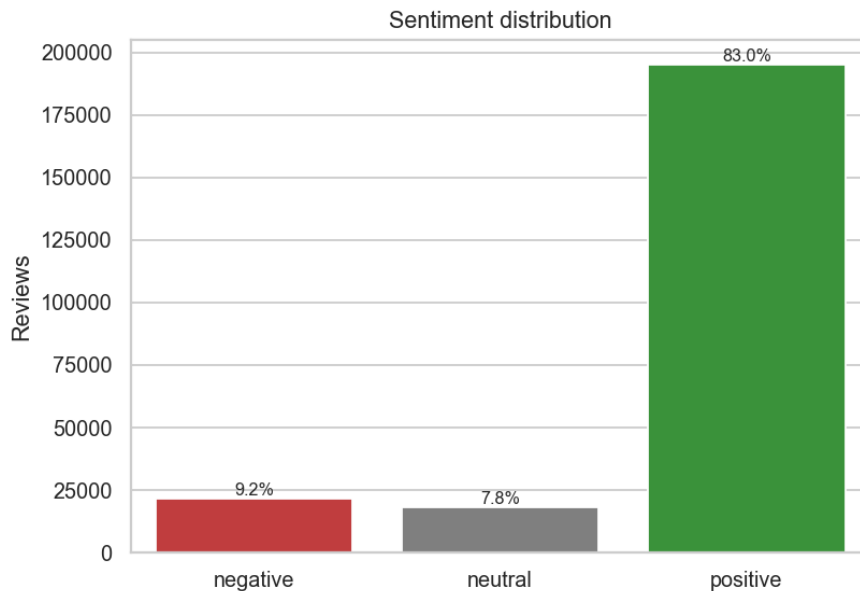


Fig 2. Distribution of derived sentiment labels.

3.2 Review length by sentiment

Neutral reviews are the longest on average (~103 words), versus ~79 for positive and ~68 for negative reviews — ambivalent reviewers write more to justify a middling verdict.

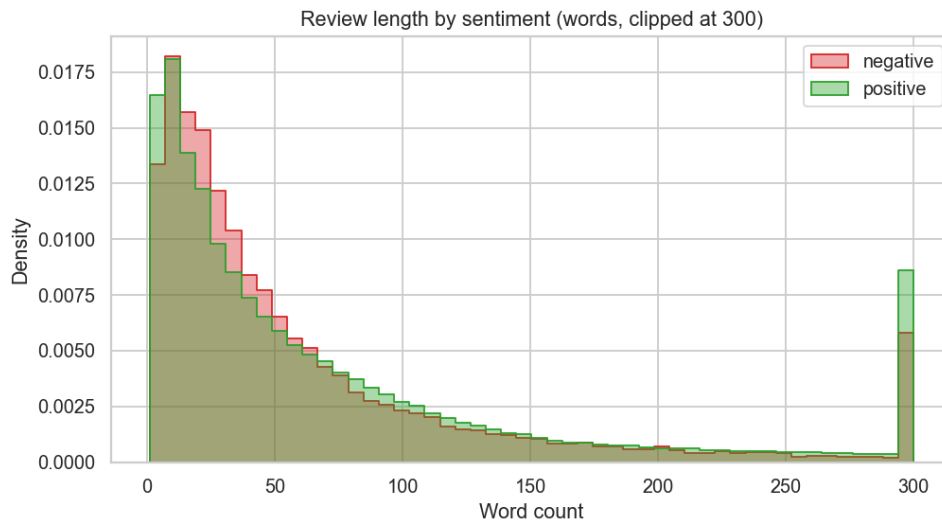


Fig 3. Review length distribution by sentiment class.

3.3 Sentiment by product category

Negative-review rates vary by category: **Electronics** is the most negative (12.9% negative) while **Books** is the least (6.6%). This points the quality team to where complaints concentrate.

Category	Mean rating	% negative
Books	4.36	6.6%
Clothing Shoes and Jewelry	4.33	9.3%
Electronics	4.23	12.9%
Home and Kitchen	4.37	9.4%
Toys and Games	4.44	7.7%

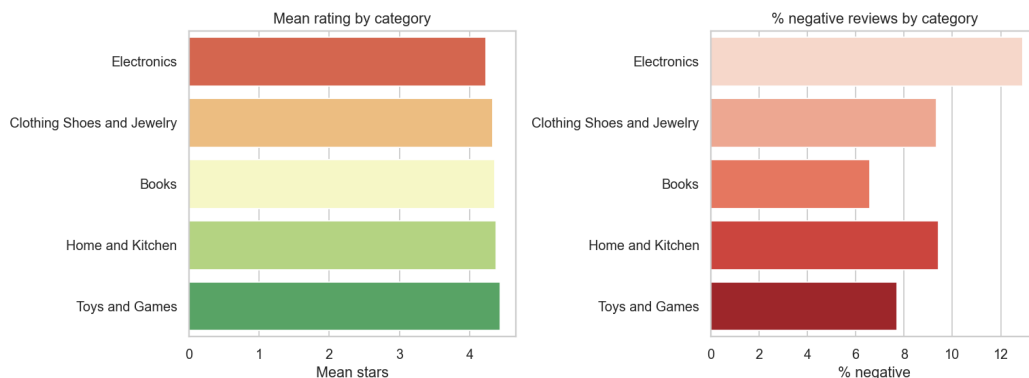


Fig 4. Sentiment split across product categories.

3.4 What words drive sentiment?

The most frequent negative bigrams cluster around product failure and regret: *waste money*, *year old*, *looks like*, *doesn't work*, *didn't work*, *don't know*, *stopped working*, *did work*. Phrases like *waste money*, *stopped working* and *poor quality* flag defects and durability issues.

Positive reviews emphasise satisfaction and recommendation: *year old*, *highly recommend*, *br book*, *really like*, *easy use*, *works great*.



Fig 5. Word clouds by sentiment class.

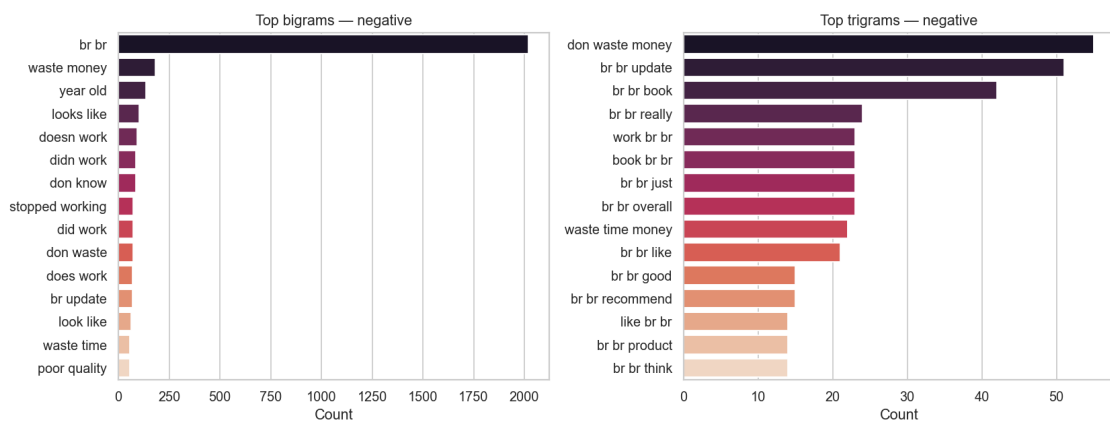


Fig 6. Most frequent n-grams by sentiment.

4. Preprocessing & Feature Engineering

Raw review text is normalised before modeling: lower-casing, stripping HTML / punctuation, removing stop-words and lemmatising to a clean token string. Example:

Raw	The protective film does not cover the entire surface of the AirTag. However, I do like the clear and clean look
Cleaned	protective film cover entire surface airtag however like clear clean look case day far good

4.1 Engineered text features

Beyond the cleaned text, lightweight numeric features capture stylistic signals. Their averages differ markedly by sentiment class:

Feature	Negative	Neutral	Positive
Avg. word count	68	103	79
Avg. character count	369	564	435
Avg. exclamation marks	0.39	0.27	0.58
Avg. ALL-CAPS words	0.84	0.76	0.57
Mean VADER score	0.001	0.373	0.723
Mean helpful votes	2.29	1.82	1.67

The unsupervised VADER score already separates the classes (negative 0.00 → neutral 0.37 → positive 0.72), confirming the labels are learnable from text alone.

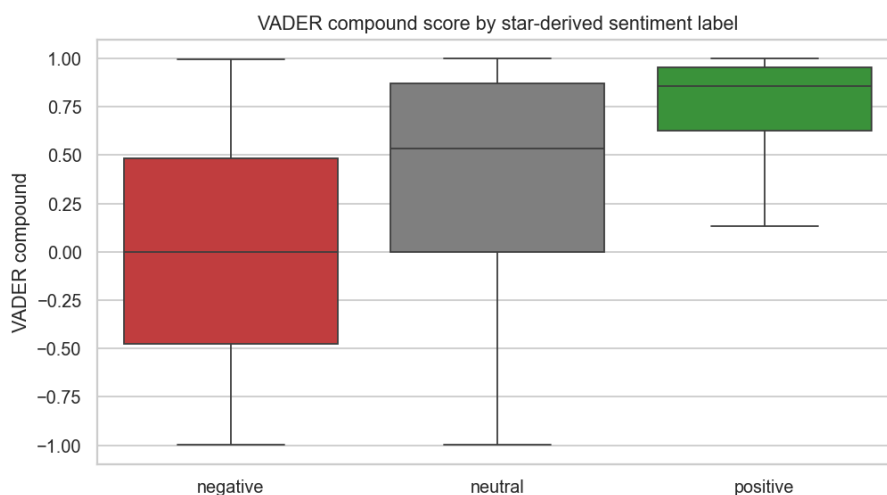


Fig 7. VADER compound score by sentiment class.

5. Modeling Approach

Two complementary models were trained and evaluated on the **same stratified held-out test set** for a fair comparison.

5.1 TF-IDF + Logistic Regression (baseline)

Reviews are vectorised with TF-IDF (50,000 features) and classified with multinomial Logistic Regression. Hyper-parameters were tuned by cross-validation (~119s of search).

Best parameters: **C** = 1.0, **class_weight** = balanced. Balanced class weights directly counter the positive skew.

5.2 DistilBERT (transformer)

A pre-trained **DistilBERT** model was fine-tuned for 3-class sequence classification (negative / neutral / positive). As a contextual transformer it captures word order and negation that a bag-of-words model cannot, at higher compute cost. The fine-tuned model is saved under `models/distilbert/` and its predictions on the shared test split are summarised by the confusion matrix in the next section.

6. Results & Model Comparison

Model	Accuracy	Macro-F1	Weighted-F1	ROC-AUC
TF-IDF + LogReg	0.798	0.605	0.824	0.888

All models are scored on the same stratified test set (n = 46,997). Macro-F1 and ROC-AUC are emphasised because accuracy is inflated by the dominant positive class.

6.1 Baseline per-class performance

Class	F1-score
Negative	0.590
Neutral	0.330
Positive	0.896

The baseline classifies positive reviews very well (F1 = 0.90) and negative reviews reasonably (F1 = 0.59), but struggles on the rare, ambiguous neutral class (F1 = 0.33) — the hardest 3★ reviews sit between classes.

6.2 Confusion matrices

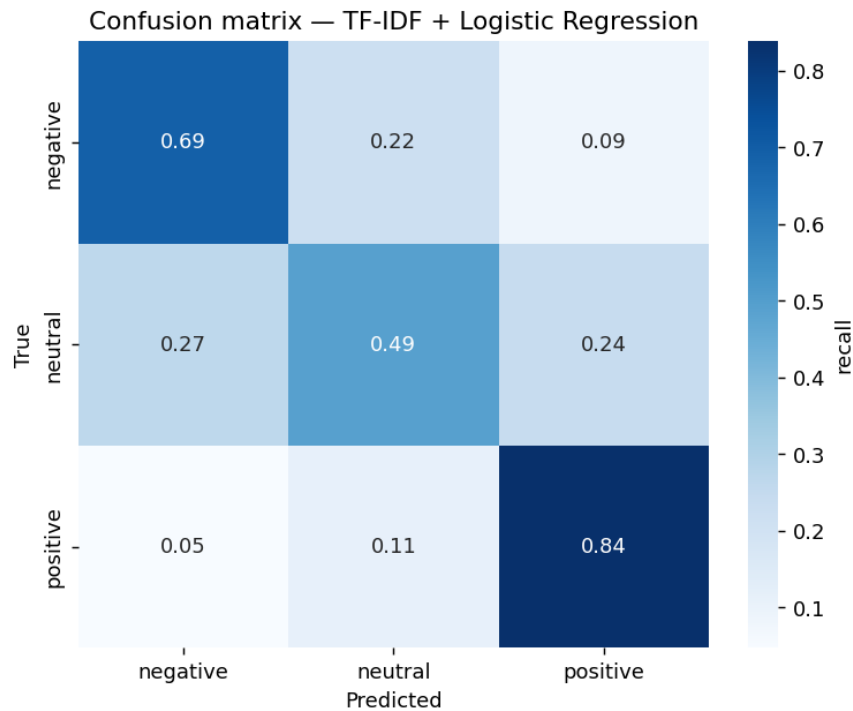


Fig 8. Confusion matrix — TF-IDF + Logistic Regression on the test set.

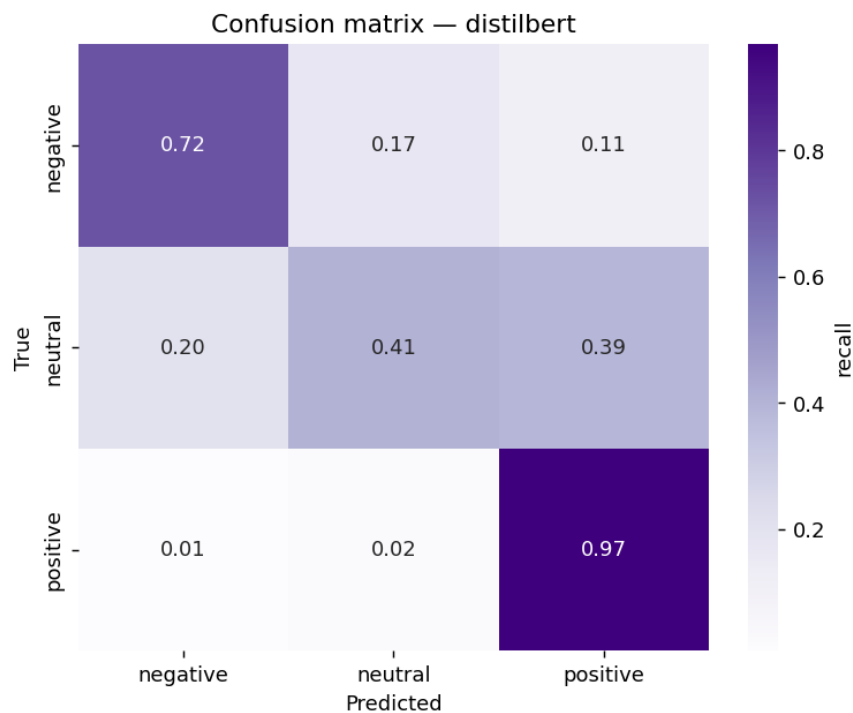


Fig 9. Confusion matrix — fine-tuned DistilBERT on the same test set.

Both models confuse neutral reviews with the adjacent positive and negative classes most often — consistent with the inherent ambiguity of 3★ reviews.

7. Business Findings & Recommendations

Key findings

- Sentiment is strongly positive: 83.0% of reviews are positive versus 9.2% negative.

- **Electronics** carries the highest negative-review rate (12.9%) and should be prioritised for quality review.
- Negative reviews are dominated by failure/regret themes (*waste money, year old, looks like, doesn't work*), pointing to durability and value complaints.
- Even a lightweight baseline reaches macro-F1 0.61 and ROC-AUC 0.89 — accurate enough to automate first-pass review monitoring.

Recommendations

- **Automate monitoring** by deploying the model to flag incoming negative reviews in real time for the support and quality teams.
- **Prioritise high-negativity categories and products** for defect investigation, using negative n-gram themes as a diagnostic starting point.
- **Promote DistilBERT for production scoring** where context and negation matter; keep the TF-IDF baseline as a fast, interpretable fallback and sanity check.
- **Watch the neutral class** — route low-confidence / 3★ predictions to human review rather than auto-actioning them.

8. Future Work

- **Aspect-based sentiment** to attribute complaints to specific product attributes (battery, sizing, durability).
- **Larger / domain-adapted transformers** (e.g. RoBERTa, full BERT) and threshold tuning to lift neutral-class recall.
- **Explainability at scale** — token-attribution and SHAP summaries surfaced directly in the monitoring dashboard.
- **Product-level dashboards** ranking the worst products by negative rate, with drill-down into representative reviews.
- **Deployment** behind an API for real-time scoring of new reviews.

Generated from project artifacts in `outputs/`. All metrics are taken directly from the pipeline's JSON outputs; figures are embedded from `outputs/figures/`.